

Fatmah Almeer

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Bilingual (English and Arabic)



EDUCATION

The Pennsylvania State University

B.S. Social Data Analytics | B.S. Hospitality Management

Minor: Entrepreneurship and Innovation

Expected May 2026

Programming Languages: Python, SQL, R

Data Engineering: ETL Pipelines, Data Modeling

Analytics & ML: Machine Learning, RAG, LLM Applications, Statistical Analysis

Tools: Flask, Alteryx, Qualtrics, PostgreSQL, Azure

Visualization: Tableau, Power BI

EXPERIENCE & PROJECTS

Research Assistant

Penn State University

August 2025 – Present

Penn State Center for Social Data Analytics

- Built and optimized end-to-end data pipelines in Python to ingest, transform, and validate large-scale multimodal datasets (1,400+ public meeting videos), improving data quality and enabling scalable analysis.
- Designed and implemented a retrieval-augmented generation (RAG) system (LlamaIndex + Llama 3) to support structured querying and insight generation from unstructured data.
- Developed and applied a structured data schema and codebook for classification tasks, improving dataset consistency and supporting downstream machine learning workflows.
- Led coordination across student contributors, ensuring data integrity, consistent annotation standards, and efficient pipeline execution.

Free Food Map Project

Penn State University

August 2023 – Present

Project Lead & Developer - Food Decisions Research Laboratory

- Designed and implemented a relational PostgreSQL database and Flask-based web application to manage and serve 500+ records, enabling efficient querying and data access.
- Developed ETL-style data pipelines for ingestion, validation, and transformation of user-submitted data, ensuring high data integrity and reliability.
- Architected backend workflows including role-based access control, admin review systems, and automated validation.
- Deployed and managed a cloud-based application on Azure, optimizing scalability, performance, and system availability.
- Led a cross-functional team to scale the dataset, improving coverage, accuracy, and usability.

Data Analytics Intern

McLean, VA

May 2024 – May 2025

Hilton

- Developed and deployed Tableau dashboards and automated reporting tools to support strategic decision-making on global hotel development and expansion.
- Performed data analysis and modeling on large datasets to evaluate market opportunities, acquisition targets, and brand expansion strategies.
- Collaborated with cross-functional stakeholders to translate business requirements into structured analytical solutions and reporting frameworks.
- Designed and standardized a global reporting pipeline, improving efficiency, consistency, and scalability of internal analytics processes.

PUBLICATION

From Miso to Tray-C: The Impact of Hedonic and Utilitarian Name Preference on Non-Humanoid Service Robots in Restaurants DOI: <https://doi.org/10.61701/843459.076>

ICHRIE Research Reports, January 2026